Gas with variable heat capacity — Solution

W26 (2026) — English translation

A1 **0.50 pts**

Determine the dependence of the internal energy of such a gas on temperature. Assume that at $T = 0$, the internal energy is zero.

Let us write down the infinitesimal increment of the internal energy of the gas:

$$\mathrm{d}U = C_V \mathrm{d}T = (C_0 + aT)\mathrm{d}T$$

Integrating, we obtain:

Answer:

$$U = C_0 T + \frac{aT^2}{2}$$

A2 **0.30 pts**

What is the heat capacity of the gas C_P at constant pressure at a given temperature T?

According to Mayer's relation:

$$C_P = C_V + R \cdot 1 \text{ mol}$$

Here and below, we omit the 1 mol factor in all expressions. From this, we get:

Answer:

$$C_P = C_0 + aT + R$$

A3 **0.40 pts**

Consider an isochoric process where the pressure of a gas increases from p_1 to p_2 at a constant volume V. Find the amount of heat supplied, Q_V.

According to the First Law of Thermodynamics:

$$\mathrm{d}Q = \mathrm{d}U + \mathrm{d}A$$

Since the process is isochoric, the gas does no work, and

$$Q_V = \Delta U = C_0 \Delta T + \frac{a(T_2^2 - T_1^2)}{2}$$

Using the ideal gas equation of state:

$$pV = RT,$$

we can find the initial and final temperatures of the process:

$$T_1 = \frac{p_1 V}{R}$$

$$T_2 = \frac{p_2 V}{R}$$

Expressing the answer in terms of the given quantities:

Answer:

$$Q_V = \frac{C_0(p_2 - p_1)V}{R} + \frac{a(p_2^2 - p_1^2)V^2}{2R^2}$$

A4 — 0.40 pts

Consider an isobaric process where the volume of a gas increases from V_1 to V_2 at a constant pressure p. Find the amount of heat supplied, Q_P.

Let us write the infinitesimal increment of heat:

$$\mathrm{d}Q = C_P \mathrm{d}T = (C_0 + aT + R)\mathrm{d}T$$

By integrating, we obtain a result similar to the previous section:

$$Q_P = (C_0 + R)\Delta T + \frac{a(T_2^2 - T_1^2)}{2}$$

Using the equation of state, we find the initial and final temperatures of the process:

$$T_1 = \frac{pV_1}{R}$$

$$T_2 = \frac{pV_2}{R}$$

Expressing the answer in terms of the given quantities:

Answer:

$$Q_P = \frac{(C_0 + R)(V_2 - V_1)p}{R} + \frac{a(V_2^2 - V_1^2)p^2}{2R^2}$$

A5 — 1.50 pts

Consider a cycle 123, consisting of a process $1-2$, in which the pressure is directly proportional to the volume, an isochore $2-3$, and an isobar $3-1$. It is given that $V_2 = 2V_1$, and the heat capacities at constant volume at states 1 and 2 are related by $C_{V2} = 1.5C_{V1}$. Determine the efficiency of this cycle. Find the numerical value. Assume $C_0 = 3R/2$.

Equation of state:

$$pV = RT$$

In process $1-2$, $p = \alpha V$, which implies $T \propto V^2$. Therefore, $T_2 = 4T_1$, $p_2 = 2p_1$, and $\alpha V_1^2 = RT_1$. Using the given condition for the heat capacities:

$$C_0 + aT_2 = C_0 + 4aT_1 = 1.5(C_0 + aT_1),$$

from which it follows:

$$C_0 = 5aT_1$$

Heat is supplied only during process $1-2$ (where the temperature monotonically increases), and is rejected only during process $3-1$ (where the temperature monotonically decreases). Let's find the supplied heat:

$$\mathrm{d}Q = C_V \mathrm{d}T + p\mathrm{d}V = (C_0 + aT)\mathrm{d}T + \alpha V \mathrm{d}V$$

After integration, we get:

$$Q_+ = C_0 \cdot (4T_1 - T_1) + a \cdot \frac{16T_1^2 - T_1^2}{2} + \alpha \cdot \frac{4V_1^2 - V_1^2}{2} = \frac{9}{2}C_0T_1 + \frac{3}{2}RT_1 = \frac{33}{4}RT_1$$

Now we find the work done by the gas: for process $1-2$, $A_{12} = \frac{3}{2}RT_1$; process $2-3$ is isochoric, so $A_{23} = 0$; for the final process at pressure p_1, the volume changes from $2V_1$ to V_1, so $A_{31} = -p_1V_1 = -RT_1$. The total work done by the gas is $A = \frac{RT_1}{2}$, and the efficiency is calculated as:

$$\eta = \frac{A}{Q_+} = \frac{2}{33}$$

Answer:

$$\eta = \frac{2}{33}$$

A6 — 0.50 pts

Consider an adiabatic process performed with a gas. Derive the relationship between infinitesimal increments of volume $\mathrm{d}V$ *and temperature* $\mathrm{d}T$*. Express your answer using* C_0*,* a*,* T*,* V*, and* R*.*

For an adiabatic process, the infinitesimal heat transfer is zero:

$$\mathrm{d}Q = C_V \mathrm{d}T + p\mathrm{d}V = 0$$

Using the equation of state:

$$p = \frac{RT}{V}$$

Substituting this into the equation, we have:

$$(C_0 + aT)\mathrm{d}T + \frac{RT}{V}\mathrm{d}V = 0$$

Answer:

$$(C_0 + aT)\mathrm{d}T + \frac{RT}{V}\mathrm{d}V = 0$$

A7 — 0.90 pts

From the relation in the previous section, obtain an adiabatic equation connecting T *and* V *of the form*

$$f(T, V) = \mathrm{const},$$

where the function f *may also include the parameters* C_0*,* a*, and* R*.*

Dividing the expression from the previous section by T, we get:

$$\left(\frac{C_0}{T} + a\right)\mathrm{d}T + \frac{R}{V}\mathrm{d}V = 0$$

Integrating, we get:

$$f(T, V) = C_0 \ln \frac{T}{T_0} + R \ln \frac{V}{V_0} + aT = \mathrm{const}$$

Here, T_0 and V_0 are arbitrary constants that ensure dimensionless arguments for the logarithms. Omitting this formality, we can present the final equation as:

Answer:

$$f(T, V) = C_0 \ln T + R \ln V + aT = \mathrm{const}$$

A8 — 0.40 pts

Consider the Carnot cycle 1234*, consisting of isotherms* 12 *and* 34 *and adiabats* 23 *and* 41*. Express the volumes* V_3 *and* V_4 *in terms of* V_1*,* V_2*, the temperatures of the isotherms* $T_+ = T_1 = T_2$ *and* $T_- = T_3 = T_4$*, and the gas parameters* C_0*,* a*, and* R*.*

Applying the equation from the previous section to the adiabat $2 - 3$:

$$C_0 \ln T_+ + R \ln V_2 + aT_+ = C_0 \ln T_- + R \ln V_3 + aT_-,$$

which transforms into:

$$R \ln \frac{V_3}{V_2} = C_0 \ln \frac{T_+}{T_-} + a(T_+ - T_-),$$

from which it follows:

$$V_3 = V_2 \left(\frac{T_+}{T_-}\right)^{\frac{C_0}{R}} \exp\left[\frac{a(T_+ - T_-)}{R}\right]$$

Similarly, we write for the adiabat $4 - 1$:

$$C_0 \ln T_- + R \ln V_4 + aT_- = C_0 \ln T_+ + R \ln V_1 + aT_+,$$

which transforms into:

$$V_4 = V_1 \left(\frac{T_+}{T_-}\right)^{\frac{C_0}{R}} \exp\left[\frac{a(T_+ - T_-)}{R}\right]$$

Answer:

$$V_3 = V_2 \left(\frac{T_+}{T_-}\right)^{\frac{C_0}{R}} \exp\left[\frac{a(T_+ - T_-)}{R}\right]$$

$$V_4 = V_1 \left(\frac{T_+}{T_-}\right)^{\frac{C_0}{R}} \exp\left[\frac{a(T_+ - T_-)}{R}\right]$$

A9 — 0.80 pts

By direct calculation (without using the general formula), show that the efficiency of this Carnot cycle is the same as that for an ideal gas with constant heat capacity.

Heat is supplied and rejected only during the isothermal processes (supplied during $1-2$, rejected during $3-4$).

$$\mathrm{d}Q = \mathrm{d}U + \mathrm{d}A = \mathrm{d}A = p\mathrm{d}V$$

(Since the process is isothermal, $\mathrm{d}U = C_V \mathrm{d}T = 0$). Using

$$pV = RT,$$

we get

$$\mathrm{d}Q_+ = RT_+ \frac{\mathrm{d}V}{V},$$

$$Q_+ = RT_+ \ln\frac{V_2}{V_1}.$$

Similarly:

$$Q_- = RT_- \ln\frac{V_4}{V_3}$$

From the previous section, we know:

$$\frac{V_3}{V_4} = \frac{V_2}{V_1}$$

Let us express the efficiency:

$$\eta_1 = 1 - \frac{|Q_-|}{Q_+} = 1 - \frac{T_-}{T_+}$$

The standard efficiency of a Carnot cycle is:

$$\eta_c = 1 - \frac{T_-}{T_+},$$

which confirms that the expressions for the efficiency coincide.

Answer: The efficiency of the cycle coincides with the efficiency of the standard Carnot cycle.

B1 — 1.00 pts

Find the amount of heat that needs to be supplied to the gas during isochoric heating Q_V. Express your answer in terms of T_1 and R.

On the linear segment, $C_V = C_0 + aT$. We obtain a system of linear equations:

$$\begin{cases} C_0 + aT_1 = 3R/2 \\ C_0 + 2aT_1 = 5R/2 \end{cases}$$

From this system, it follows:

$$aT_1 = R$$

$$C_0 = R/2$$

Let us find Q_V:

$$Q_V = \int\limits_{T_1/2}^{3T_1} C_V \mathrm{d}T = \frac{3R}{2}\left(T_1 - \frac{T_1}{2}\right) + \frac{5R}{2}\left(3T_1 - 2T_1\right) + \int\limits_{T_1}^{2T_1}\left(\frac{R}{2} + \frac{RT}{T_1}\right)\mathrm{d}T$$

$$Q_V = \frac{3RT_1}{4} + \frac{5RT_1}{2} + 2RT_1 = \frac{21}{4}RT_1$$

Answer:

$$Q_V = \frac{21}{4}RT_1$$

B2 — 0.50 pts

Find the amount of heat that must be supplied to the gas during isobaric heating Q_P. Express your answer in terms of T_1 and R.

$$C_P = C_V + R,$$

therefore:

$$Q_P = Q_V + R\Delta T = \frac{21}{4}RT_1 + \frac{5}{2}RT_1 = \frac{31}{4}RT_1$$

Answer:

$$Q_P = \frac{31}{4}RT_1$$

B3 — 1.80 pts

Let the initial volume of the gas be V_0. What will the final volume be if the temperature is increased adiabatically? Find the work done in this process. Express your answers using T_1, R, and V_0.

For an adiabatic process where the heat capacity depends linearly on temperature as $C_V = C_0 + aT$, if the temperature changes from T_{1x} to T_{2x} and the volume from V_1 to V_2, the initial and final volumes are related as follows:

$$V_2 = V_1\left(\frac{T_{1x}}{T_{2x}}\right)^{\frac{C_0}{R}} \exp\left[\frac{a(T_{1x} - T_{2x})}{R}\right]$$

This process can be divided into three segments with the following parameters:

$$C_0 = 3R/2,\ a = 0,\ T_{1x} = T_1/2,\ T_{2x} = T_1$$

$$C_0 = R/2,\ aT_1 = R,\ T_{1x} = T_1,\ T_{2x} = 2T_1$$

$$C_0 = 5R/2,\ a = 0,\ T_{1x} = 2T_1,\ T_{2x} = 3T_1$$

Then at the end of segment 1:

$$V_1 = \frac{V_0}{2\sqrt{2}}$$

At the end of segment 2:

$$V_2 = \frac{V_1}{e\sqrt{2}} = \frac{V_0}{4e}$$

At the end of segment 3:

$$V_3 = V = \frac{4\sqrt{2}V_2}{9\sqrt{3}} = \frac{V_0\sqrt{2}}{9e\sqrt{3}}$$

Let's find the work done by the gas in this adiabatic process:

$$A = -\Delta U = -Q_V = -\frac{21}{4}RT_1$$

Answer:

$$V = \frac{V_0\sqrt{2}}{9e\sqrt{3}}$$

$$A = -\frac{21}{4}RT_1$$